

Week 7 & Week 8 Milestone

Screen Time Analysis Dashboard – Report

1. Introduction

With the rapid growth of digital technology, screen usage among children has increased significantly. This study analyzes screen time behavior among Indian children aged 7 to 18 years, focusing on:

- Age and gender differences
- Urban vs rural usage
- Device preferences
- Educational vs recreational usage
- Associated health impacts

The goal is to understand behavioral trends and highlight potential health risks.

2. Dataset Description

The dataset includes:

- Age
- Age Band (7–10, 11–14, 15–18)
- Gender
- Average Daily Screen Time (hours)

- Urban or Rural location
- Primary Device used
- Educational and Recreational screen time
- Health impact indicators (Anxiety, Eye Strain, Obesity Risk, Poor Sleep)

3. Average Screen Time by Age Band

Age Group	Avg Screen Time (hrs/day)
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7–10 yrs	3.93 hrs
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11–14 yrs	4.52 hrs
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15–18 yrs	4.51 hrs
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Interpretation:

- Screen time increases sharply after age 10.
- Teenagers maintain the highest digital exposure.
- Early adolescence is a critical phase for habit formation.

4. Screen Time by Gender

- Males have slightly higher screen time than females across all age groups.
- Both genders show:
 - Lowest usage in 7–10 years
 - Peak usage in 15–18 years

Conclusion:

Gender difference is minor but consistently favors higher male usage.

5. Urban vs Rural Screen Time

- Urban children show higher median and maximum screen time than rural children.
- Urban upper extremes reach **13–14 hours/day**, while rural peaks remain slightly lower.

Conclusion:

Better internet access, device availability, and lifestyle differences increase urban screen exposure.

6. Health Impact of Screen Time

From bubble visualization:

- **Poor Sleep – Highest impact**
- **Eye Strain – Second highest**
- **Obesity Risk – Moderate**
- **Anxiety – Lowest but noticeable**

Interpretation:

Extended screen exposure leads primarily to:

- Sleep disruption
- Visual fatigue
- Reduced physical activity

7. Educational vs Recreational Usage

Age Group	Educational Use	Recreational Use
7–10 yrs	Higher than recreation	
11–14 yrs	Nearly balanced	
15–18 yrs	Recreation dominates	

Conclusion:

As age increases:

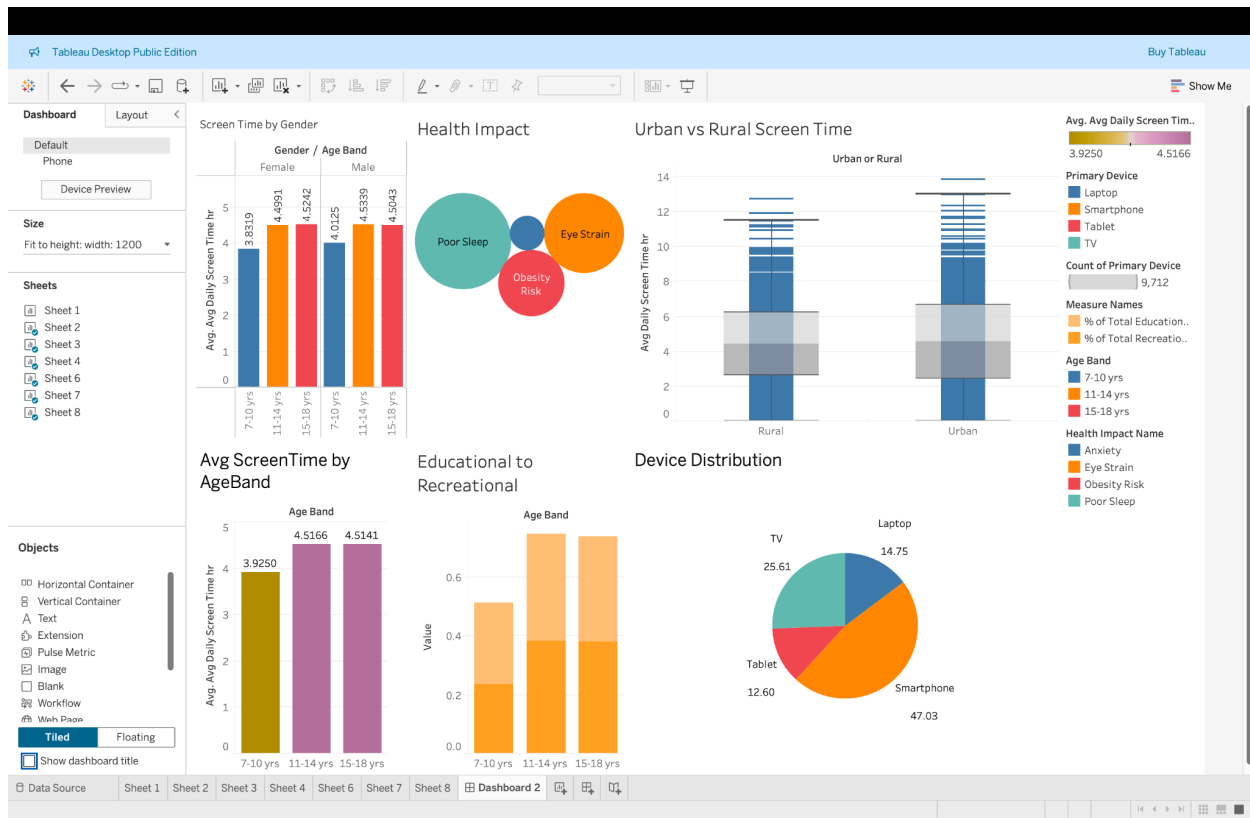
- Digital use shifts from learning to entertainment.
- Recreational dominance increases addiction risk.

8. Device Distribution

Device	Share (%)
Smartphone	47.03 %
TV	25.61 %
Laptop	14.75 %
Tablet	12.60 %

Conclusion:

Smartphones are the **primary driver** of increasing screen exposure.



9. Key Insights

- Screen time increases with age.
- Males use screens more than females.
- Urban children have higher screen exposure.
- Smartphones dominate device usage.
- Poor sleep and eye strain are the most severe health issues.
- Recreational usage overtakes educational usage with age.

10. Recommendations

1. Limit screen time for children below 14 years.
2. Encourage outdoor and physical activities.
3. Promote educational apps over gaming and social media.
4. Enable blue-light filters on devices.
5. Introduce school awareness programs on digital health.